

• Homework 3a

Consider a spacecraft with an attitude control system composed of 4 reaction wheels in an ortho-stew symmetric array. The inertia matrix of the entire spacecraft is

$$\underline{J}_c^{(B)} = \begin{bmatrix} 1200 & 100 & -200 \\ 100 & 2200 & 300 \\ -200 & 300 & 3100 \end{bmatrix} \text{ Kg m}^2$$

whereas the principal inertia moments of the rotors are

$$I_r^{(j)} = 80 \text{ Kg m}^2 \quad I_r^{(j)} = 40 \text{ Kg m}^2 \quad (j=1, \dots, 4)$$

Initial rotor speeds

$$\omega_{s1}(0) = \omega_{s2}(0) = 0.5 \text{ sec}^{-1} \quad \omega_{s3}(0) = \omega_{s4}(0) = -0.5 \text{ sec}^{-1}$$

Initial attitude

$$\underline{q}(0) = [0.1 \quad -0.2 \quad 0.5]^T \quad q_4(0) = \sqrt{1 - [q_1^2(0) + q_2^2(0) + q_3^2(0)]}$$

Initial angular rate

$$\underline{\omega}(0) = [-0.05 \quad 0.03 \quad 0.05]^T \text{ sec}^{-1}$$

The spacecraft is placed in a Molniya orbit, with orbit elements

$$a = 26554 \text{ km} \quad e = 0.73 \quad i = 63.4 \text{ deg} \quad \Omega = 0 \text{ deg} \quad \omega = -90 \text{ deg}$$

Objective is tracking the desired attitude, i.e.

$$\hat{b}_{c1} = \hat{\theta} \quad \hat{b}_{c2} = -\hat{\phi} \quad \hat{b}_{c3} = -\hat{\psi}$$

The initial true anomaly equals 90 deg.

For the purpose of obtaining the gain matrices A and B , use the following parameters for ξ and ω_n :

$$\omega_n = 0.04 \text{ sec}^{-1} \quad \text{and} \quad \xi = 1$$

These are the objectives, split in part (A) and (B)

(A) Neglect the actuator dynamics; obtain the nonlinear control law after finding A and B and apply it in the numerical integration of the attitude dynamics, in the interval $0 \leq t \leq 600 \text{ sec}$. Portray the time histories of

- (i) error quaternion, i.e. $\underline{q}_e(t)$ and $q_{e0}(t)$
- (ii) error angular rate, i.e. $\underline{\omega}_0(t)$
- (iii) commanded torque components $\underline{T}_c(t)$

(B) Consider the actuator dynamics, i.e. the skew array of RWs, together with the nonlinear control law. Integrate the attitude dynamics in $0 \leq t \leq 600 \text{ sec}$. Portray the time histories of

- (i) error quaternions, i.e. $\underline{q}_e(t)$ and $q_{e0}(t)$
- (ii) error angular rate, i.e. $\underline{\omega}_0(t)$
- (iii) commanded torque components, $\underline{T}_c(t)$
- (iv) actual torque components, $\underline{T}_a(t)$
- (v) rotor angular rates $\{\omega_{s1}(t), \dots, \omega_{s4}(t)\}$